

Final Project Submission

Futureense AI Clinic | Capstone Project 05
Client persona: Northstar Analytics

1. Project Title

Northstar Autonomous Analytics and Insight Copilot

Agentic analytics system: plain-language business questions over Olist e-commerce data, producing verified SQL and plain-language insight with query exposure.
Short name: Northstar Insight Copilot

2. Names of All Team Members

#	Name	Git identity	Primary focus
1	Ankit	ankit-2244	Product & discovery (Sprint 0)
2	Vishal	vicodwer	Design, data, core pipeline (Sprints 1-2)
3	Shanmuk	ShanmukYadav	Scale, gateway, operate & present (Sprints 3-4)

Squad size: 3
Contributions follow git commit history on main (authors ankit-2244, vicodwer, ShanmukYadav).

3. Sprint Updates

Sprint	Focus	Status	Led in git
Sprint 0	Discover & Define	Complete	Ankit
Sprint 1	Design & De-risk	Complete	Vishal
Sprint 2	Build the Core	Complete	Vishal
Sprint 3	Harden, Scale, Optimize	Complete	Shanmuk
Sprint 4	Verify, Operate, Present	Complete	Shanmuk

Sprint 0 - Discover & Define (Ankit)

- Discovery brief, personas, probing questions
 - PRD v1: problem, scope, metrics, NFRs
 - Evaluation plan, risk register v1, team charter
- Evidence: docs/prd.md, docs/stage1_discover/, docs/stage2_define_evalplan/, docs/stage4_risk/, docs/team_charter.md

Sprint 1 - Design & De-risk (Vishal)

- Architecture, orchestration decision, agent contracts, verifier contract
 - DuckDB sandbox, synthetic data pipeline, golden set seed, eval harness
 - Riskiest-assumption spike (Verifier)
- Evidence: docs/stage3_design/, src/sandbox/, src/synthetic/, src/spike/, evals/

Sprint 2 - Build the Core (Vishal)

- Live agents: Router, Query Writer, Verifier, Narrator
 - End-to-end pipeline with retry-once
 - First eval scripts (router, query writer, full pipeline)
- Evidence: src/agents/, src/verifier/, src/pipeline.py, src/eval/eval_*.py

Sprint 3 - Harden, Scale, Optimize (Shanmuk)

- LLM gateway, cache, agent registry
- Clarifier + Planner agents
- Semantic result matching, regression tests
- A/B narrator harness, burst load, DSPy/prompt scaffold

Evidence: `src/gateway/`, `src/registry/`, `clarifier/planner`, `tests/`, `eval harnesses`

Sprint 4 - Verify, Operate, Present (Shanmuk)

- FastAPI + web UI
- Ops runbook, SLOs, canary/rollback
- Baseline benchmark vs single-LLM path
- Presentation script, viva prep, close-out docs

Evidence: `src/api/`, `ops/`, `docs/sprint4/`, `baseline_benchmark.py`

Quality snapshot (system)

Metric	Value
Pipeline accuracy (golden set)	18/18 (100%) semantic match
Avg cost / question	~\$0.003
Latency p50 / p95 (sequential)	~5.5s / 8.4s
Burst (10 concurrent)	10/10 answered

4. Latest Project Status

Dimension	Status
Overall	Sprints 0-4 complete
Demo	Local FastAPI UI working (<code>uvicorn src.api.app:app --port 8000</code>)
Core quality	Golden-set 18/18; clarify & refuse working
Data	Olist in DuckDB (read-only); rebuild via <code>build_db.py</code>
Cloud	Docker/AWS pilot planned (college budget <= Rs 2000)
Presentation	Script ready: <code>docs/sprint4/presentation_20min.md</code>

Runtime architecture

User (UI/API) -> FastAPI -> LLM Gateway -> Router -> (refuse | Clarifier | Planner | standard) -> Query Writer -> Verifier (retry-once) -> Narrator -> Answer + SQL + cost/latency

How to run

1. Clone the GitHub repository
2. Create `.env` with `OPENROUTER_API_KEY=...`
3. `pip install -r requirements.txt`
4. `python src/sandbox/build_db.py` (if `sandbox.duckdb` missing)
5. `uvicorn src.api.app:app --host 0.0.0.0 --port 8000`
6. Open `http://localhost:8000/`

5. Individual Contribution of Each Team Member

Based on `git authors` and `files` in each `sprint commit`.

Ankit (ankit-2244)

- Product/discovery: discovery brief, personas, problem framing
- PRD: problem, users, scope, metrics, NFRs
- Planning: evaluation plan, risk register v1, team charter
- Sprint ownership: Sprint 0 commit

Vishal (vicodwer)

- Architecture docs, orchestration decision, agent/verifier contracts
- Data: sandbox builder, synthetic generators, golden/synthetic sets
- Core system: Router, Query Writer, Narrator, Verifier, pipeline

- Eval: eval_pipeline, eval_query_writer, eval_router
- De-risk spike; Sprint ownership: Sprints 1 and 2

Shanmuk (ShanmukYadav)

- LLM gateway, caches, agent registry
- Clarifier, Planner; semantic result_match and pytest suite
- A/B narrator, burst load, DSPy scaffold
- FastAPI API + UI; runbook, SLOs, canary/rollback
- Baseline benchmark, presentation/viva materials; Sprints 3 and 4

6. Collective / Team Contribution

- End-to-end agentic analytics copilot on multi-table Olist data
- Full lifecycle: discover -> define -> design -> risk -> data -> build -> harden -> operate
- Safety: clarify ambiguous questions; refuse destructive intent; deterministic verification
- Measured quality and cost (golden set, cost/latency, burst helper)
- Pilot operate surface: API + UI + ops docs
- Defense materials: presentation script and viva prep
- Shared standards: one repo, no secrets in git, offline unit tests

7. GitHub Repository Link

<https://github.com/ShanmukYadav/Northstar-Copilot>

Default branch: main

Repository contents checklist

Required item	Location
Source code	src/
Documentation	docs/
Datasets	Olist via rebuild / local data/ (documented)
APIs	src/api/app.py (LLM via OpenRouter)
Presentation / reports	docs/sprint4/, FINAL_SUBMISSION
Other	ops/, evals/, tests/, requirements.txt, README.md

Never commit .env or API keys.

8. Declaration

We confirm that this submission describes Sprints 0-4 on the Northstar Insight Copilot, that the GitHub repository is the source of truth for code and documentation, and that secrets are not stored in the repository.

Role	Name	Signature / Date
Team member	Ankit	_____
Team member	Vishal	_____
Team member	Shanmuk	_____

Final submission | Project 05 | Northstar Autonomous Analytics and Insight Copilot | Team of 3 (Ankit, Vishal, Shanmuk)